



Simple Linear Correlation

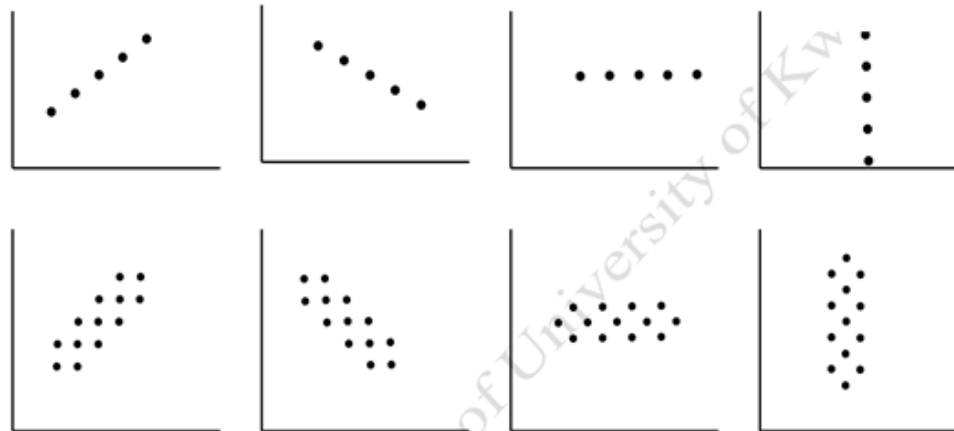
Dr Phikolomzi Matikinca

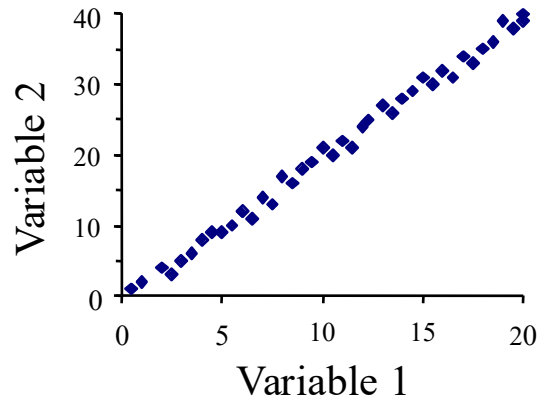
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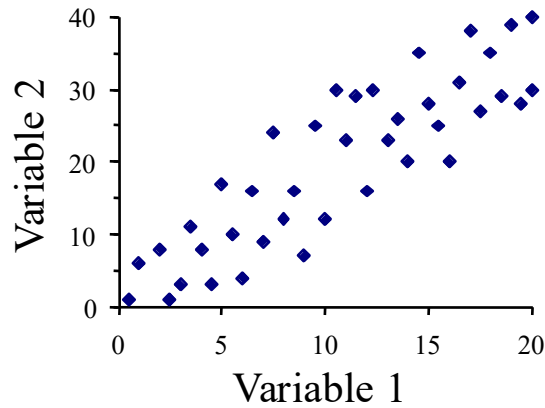
Simple Linear Correlation

- a process for establishing relationships between two **quantitative** variables (i.e. interval or ratio data)
- a statistical measure that expresses the extent to which two variables are linearly related (they change together at a constant rate)
- In other words: the extent to which two variables **covary** (vary together in a consistent way)
- **Simple:** refers to the fact that only two variables are considered
- **Linear:** only linear relationships will be considered
- “Simple” because a relationship between two variables is the simplest relationship possible

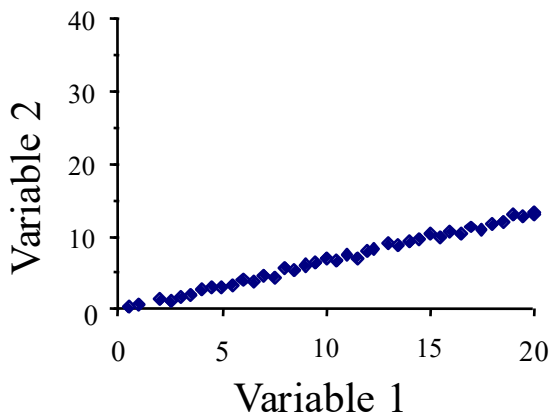




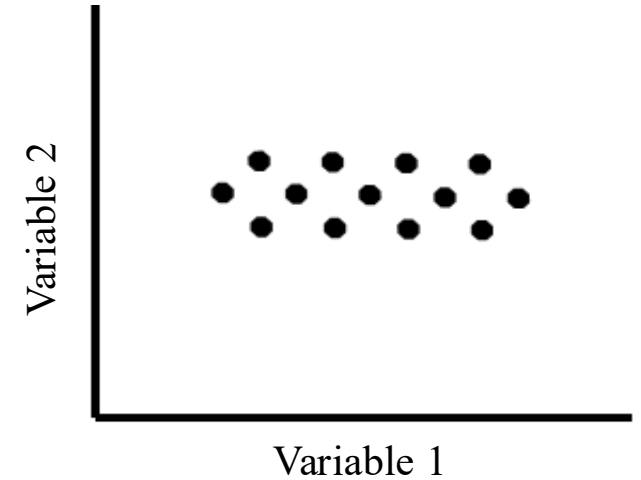
The data form a distinct line – the band of data is tight and quite filled in (i.e., there is little space between the observations). This quality makes this a **strong relationship**.



The data do NOT form a distinct line – the band of data is loose and there is quite a bit of space between the observations. This relationship is therefore **weaker** than the one above.



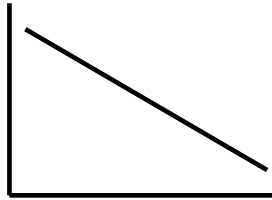
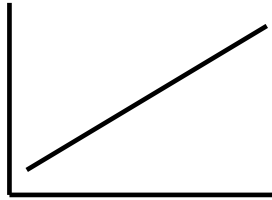
The relationship between the two variables is strong, just as in the top graph. But there's a noticeable difference between this relationship and the one in the top graph, isn't there? In this graph, changes in variable 1 have **less of an effect** on variable 2 than they do in the top graph. This means that there is a **lower magnitude** of effect of variable 1 on variable 2 in this graph than in the top graph.



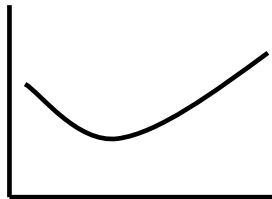
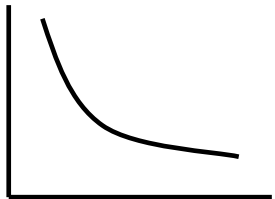
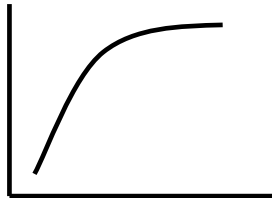
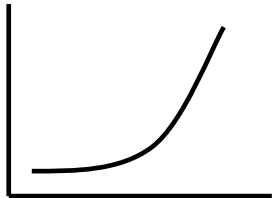
There is no relationship between var1 and var2. Var2 is not related to var2 in any way.

Simple Linear Correlation

linear



non-linear



- LINEAR – the slope of the relationship is constant
i.e., for each unit change ($\uparrow$ or $\downarrow$) in 1 variable, there is a constant change ($\uparrow$ or $\downarrow$) in the other

- A non-linear relationship doesn't have a constant slope (the shape of the relationship is not a straight line)

Simple Linear Correlation

We use a correlation index that measures
strength & direction
of a relationship between two variables

Simple Linear Correlation

Pearson correlation coefficient (r)



Karl Pearson

- Summarizes the relationship between two continuous variables in a single number called the correlation coefficient
 - Denoted by r
 - The strength of the linear relationship between two continuous variables (i.e. temperature and pH)
 - Most appropriate for normally distributed data
 - An alternative approach is the Spearman rank correlation,....
 - ...which looks only at the data ranks in the two timeseries

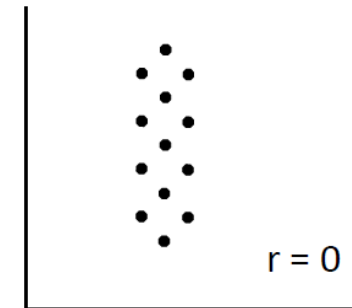
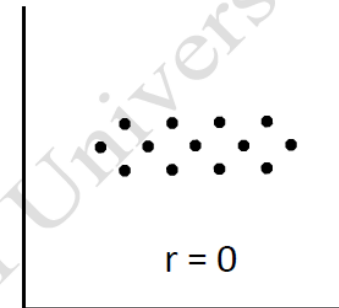
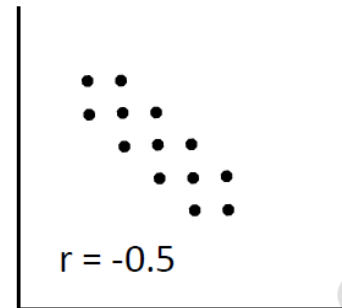
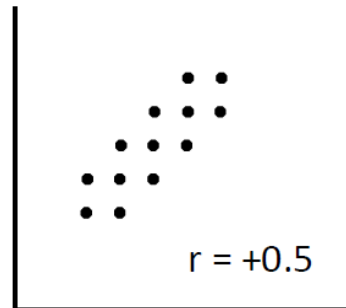
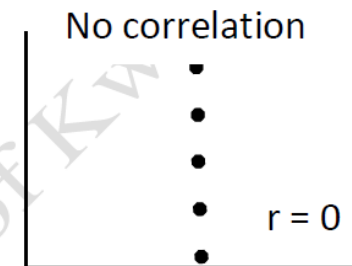
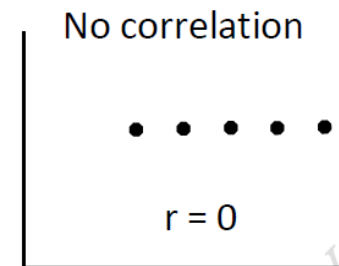
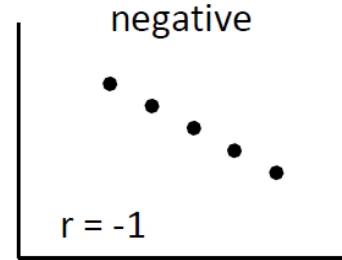
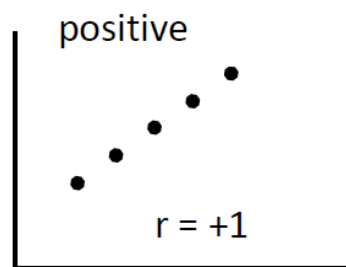
Simple Linear Correlation

Pearson correlation coefficient (r)

- Total negative linear correlation = **-1**
- No correlation = **0**
- Total positive correlation = **+1**



Karl Pearson



Simple Linear Correlation

Pearson correlation coefficient (r)

- The Pearson correlation for two objects
 - Sums the product of their differences from their object means, and divides the sum by the product of the squared differences from the object means

$$r = \frac{\sum (x_i - \bar{x}) (y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

r = correlation coefficient

x_i = values of the x-variable in a sample

$\bar{x}$ = mean of the values of the x-variable

y_i = values of the y-variable in a sample

$\bar{y}$ = mean of the values of the y-variable

Simple Linear Correlation

Pearson correlation coefficient (r)

$$r = \frac{\sum (x_i - \bar{x}) (y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

Simple Linear Correlation

Pearson correlation coefficient (r)

X_i and Y_i are the i^{th} observations of the variables x and y , and n is the total number of pairs of x - y observations

$$r = \frac{\sum_{i=1}^n X_i Y_i - \frac{\sum_{i=1}^n X_i \sum_{i=1}^n Y_i}{n}}{\sqrt{\left(\sum_{i=1}^n X_i^2 - \frac{\left(\sum_{i=1}^n X_i \right)^2}{n} \right) \left(\sum_{i=1}^n Y_i^2 - \frac{\left(\sum_{i=1}^n Y_i \right)^2}{n} \right)}}$$

➡ Extent to which the variables X and Y vary together (**covary**)

➡ Extent to which the variables X and Y vary separately

$$r = \frac{\text{Common variation of X \& Y}}{\text{Unexplained variation in X \& Y}}$$

- $\therefore$ A ratio of the common variability to the unexplained variability within each variable
- The higher the common variation and the lower the unexplained variation, the higher the correlation coefficient

Simple Linear Correlation

Coefficient of determination or correlation index (r^2)

- Square of the correlation coefficient (r^2 or R^2)
- Indicates the proportion of the variability in one variable that is accounted for by the other variable to which it is correlated
- In other words: the percentage of variation in the dependent variable explained by variation in the independent variable
- $r^2 \times 100$ gives the % of variability in the dependent variable that is explained by the variability in the independent variable to which it is correlated
- In a perfect correlation ($r = 1$, $r^2 = 1$), 100% of the variability in the dependent variable is accounted for by the variability in the independent variable.
- When $r = 0$, $r^2 = 0$, 0% of the variability in one variable is explained by the other variable

Simple Linear Correlation

How to evaluate the relationship between two variables?

- **r** is an estimator of the population correlation coefficient (ρ , the greek letter rho)
- **NB:** It's not a p
- Since it is an estimate, there is uncertainty in **r**
- This uncertainty is used in the evaluation of the significance of a calculated correlation coefficient

Simple Linear Correlation

How to evaluate the relationship between two variables?

- The significance of r may be calculated with the t -statistic used in the t -test, with t calculated as:

$$t = \frac{r}{S_r},$$

- Where S_r is the standard error of the correlation coefficient, which, is a measure of uncertainty of r .

$$S_r = \sqrt{\frac{1 - r^2}{n - 2}}$$

- Where n is the sample size.
 - It is number of x - y observations (number of pairs of observations (x_i, y_i) in the dataset)
- $n - 2$ represents degrees of freedom
- You lose **TWO** degrees of freedom because you are estimating two parameters from your data (the Y -intercept and the slope) to determine the relationship.

Simple Linear Correlation

Pearson correlation coefficient (r)

Example:

In a study to determine whether there is a relationship between bird nest abundance and altitude

$H_0: \rho = 0$

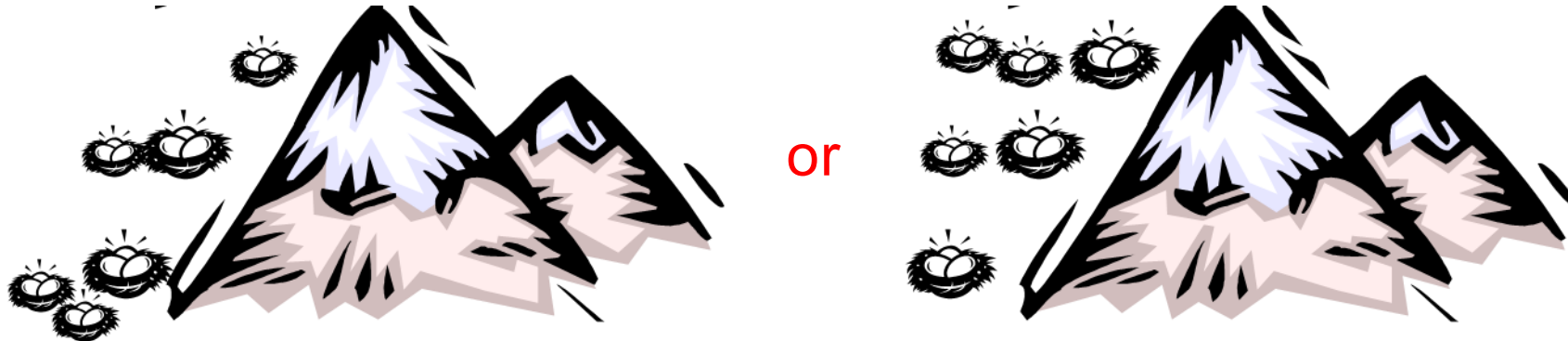
i.e., the population correlation coefficient for nest abundance and altitude = 0

Implies no linear relationship between nest abundance and altitude

$H_A: \rho \neq 0$

i.e., the population correlation coefficient $\neq 0$

Implies a linear relationship between nest abundance and altitude



Simple Linear Correlation

Pearson correlation coefficient (r)

Example:

SPSS Output

Correlations

		number_ nests	altitude
number_nests	Pearson Correlation	1	-.990**
	Sig. (2-tailed)		.000
	N	10	10
altitude	Pearson Correlation	-.990**	1
	Sig. (2-tailed)	.000	
	N	10	10

**. Correlation is significant at the 0.01 level (2-tailed)

$r = -0.990$; number of nests vs. altitude and altitude vs. number of nests are the same thing and therefore produce the same results.

$p = 0.000$, which should be written " $p < 0.0005$ " because p is not truly 0.

Statistical conclusion: number of birds' nests and altitude are significantly related to each other. The relationship is negative (as indicated by $r = -0.990$). As altitude increases, the number of birds' nests decreases.



THANK YOU